# AIMS5701: Fundamentals in AI

## Homework 4: HMMs and Particle Filtering

**Deadline: 23:59, November 9, 2026**

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## Important Regulations and Policies

### Submission Guidelines

- Please submit your declaration and solutions on Blackboard (under the "course contents" tab).
- Your submission must be a **single PDF file**. Submissions in other formats (e.g., image files, Word documents) will not be graded.
- Please write your name and student ID on the first page of your solutions.
- For the late submission policy, please refer to regulations posted on Blackboard.

### Collaboration Policy

- You are allowed to discuss the problems with your classmates to better understand the concepts.
- However, you **MUST** write up your own solutions independently.
- You **MUST NOT** copy from your classmates or from any other material. Do not share your final written solutions with others.

### Policy on AI Tools (e.g., ChatGPT, Copilot, Gemini)

- Our policy is "Approach 2 – Use only with prior permission." For this assignment, you are permitted to use AI tools to help you understand concepts.
- **Acknowledgement is mandatory.** If you use an AI tool for any part of a problem, you **MUST** attach the full, unedited transcript of your interaction with the tool for that problem. This includes the prompts you used and the original answer generated by the AI. This should be attached at the end of your submission. Failure to do so will be considered a violation of academic integrity.

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# Questions

**Instructions.** There are 6 problems and 14 subquestions. Show your calculations and justify conceptual answers. Keep exact values during intermediate calculations; give exact final probabilities or round to four decimal places. Matrix and state orderings are specified in each problem.

### Problem 1: Markov Chain Calculations

At the start of each lunch period, a campus cafeteria is either busy ($B$) or quiet ($Q$). Model its condition across successive lunch periods as a time-homogeneous Markov chain. The transition matrix is

$$T = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}, \qquad T_{ij} = P(X_{t+1} = j \mid X_t = i).$$

Both the rows (current state) and columns (next state) are ordered $(B, Q)$. Initially, $P(X_1 = B) = 1$ and $P(X_1 = Q) = 0$.

(a) Compute the probability distribution $P(X_2)$ for the second lunch period. Show the calculation for each state.

(b) Using the Mini-Forward Algorithm, compute $P(X_3)$ from your result in part (a).

(c) Compute the stationary distribution $\pi = (\pi_B, \pi_Q)$. Set up and solve the stationarity and normalization equations.

### Problem 2: HMM Filtering: The Forward Algorithm

A meeting room is either occupied ($O$) or vacant ($V$). Its occupancy $X_t$ is hidden. A motion sensor produces $E_t = +$ when it detects activity and $E_t = -$ otherwise. Use the following HMM:

$$P(X_0 = O) = 0.4, \qquad P(X_0 = V) = 0.6,$$

$$T = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}, \qquad L = \begin{pmatrix} 0.75 & 0.25 \\ 0.25 & 0.75 \end{pmatrix}.$$

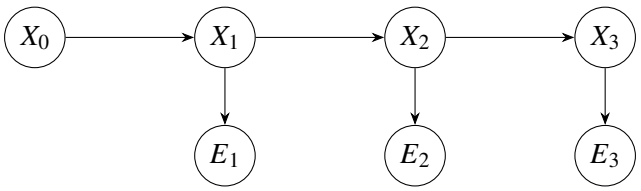
Here $T_{ij} = P(X_t = j \mid X_{t-1} = i)$, with both rows and columns ordered $(O, V)$. Also $L_{i,e} = P(E_t = e \mid X_t = i)$, with rows ordered $(O, V)$ and columns ordered $(+, -)$. The two observations are $e_1 = +$ and $e_2 = -$. The initial distribution is at time 0: first transition to $X_1$, then incorporate $e_1$.

(a) Compute $B_1(x) = P(X_1 = x \mid e_1)$ for both states. Show the prediction, observation update, and normalization.

(b) Compute $B_2(x) = P(X_2 = x \mid e_1, e_2)$ for both states, again showing the prediction, observation update, and normalization.

## Problem 3: HMM Conditional Independence

Consider a standard first-order HMM with hidden states $X_0, \ldots, X_T$ and observations $E_1, \ldots, E_T$, where $T \geq 3$. Its only directed edges are $X_{t-1} \to X_t$ and $X_t \to E_t$. The first part of the graph is shown below; the same pattern continues. Let $X_{a:b}$ denote $(X_a, \ldots, X_b)$, and use the same convention for $E_{a:b}$.




(a) State the two main conditional independence properties of this HMM: the first-order Markov property for hidden states and the observation independence property.

(b) Does the graph imply that $E_1$ and $E_2$ are independent without conditioning on any variables? Justify your answer using d-separation.

(c) Does the graph imply $E_1 \perp\!\!\!\perp E_2 \mid X_1$? Justify your answer using d-separation.

## Problem 4: Particle Filtering: The Elapse Step

An indoor delivery robot is localized on a grid. At time $t$, all 20 equally weighted particles are at $(4, 2)$. The transition probabilities from that location are

| Next location $s$ | $(4,3)$ | $(4,2)$ | $(3,2)$ |
|---|---|---|---|
| $P(X_{t+1} = s \mid X_t = (4,2))$ | 0.60 | 0.25 | 0.15 |

All other next locations have probability zero. Each particle is propagated independently according to this transition model. No new observation has yet been incorporated.

(a) Describe the Passage of Time (Elapse) step in particle filtering.

(b) Compute the expected number of particles at each of the three possible next locations after this step.

## Problem 5: Particle Filtering: Observe and Resample

In a *separate localization snapshot* for an indoor delivery robot, there are 10 equally weighted particles before the observation update. This particle set is supplied independently of Problem 4. Location labels and the likelihood of a positive beacon reading $R$ are given below. The observed reading is $r = +$.

| Location | Grid coordinate | $P(R = + \mid X = \text{location})$ |
|---|---|---|
| $A$ | $(4, 3)$ | 0.8 |
| $B$ | $(3, 2)$ | 0.5 |
| $C$ | $(4, 1)$ | 0.3 |
| $D$ | $(4, 2)$ | 0.6 |
| $E$ | $(2, 3)$ | 0.1 |
| $F$ | $(3, 3)$ | 0.3 |

The negative-reading likelihood at each location is the complement of the positive-reading likelihood. The complete particle list is:

| Particle $i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Location $S_i$ | $A$ | $B$ | $A$ | $C$ | $D$ | $E$ | $B$ | $F$ | $C$ | $A$ |

(a) **Weighting.** Calculate the unnormalized likelihood weight $w_i = P(r \mid X = S_i)$ for each of the 10 particles, and their total $W = \sum_{i=1}^{10} w_i$.

(b) **Resampling.** Suppose 10 new particles are drawn independently with replacement from the weighted particle list. For a single draw, compute the probability of selecting each of the six locations $A, B, C, D, E, F$.

## Problem 6: Conceptual Comparison

(a) When is particle filtering preferred over exact HMM filtering?

(b) How is the belief $B_t(x) = P(X_t = x \mid e_{1:t})$ represented in particle filtering?